

Batched tensor recursion for the exact discrete Fourier transform: a power saving of 607/1250000 in the logarithm

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9 October 2026

Abstract

We show that the discrete Fourier transform of every length n can be computed exactly in

$$O(n(\log n)^\theta (\log \log n)^{4-\theta}), \quad \theta = 1 - \frac{607}{1250000} = 1 - 4.856 \times 10^{-4},$$

operations, in the exact complex-arithmetic model of OpenAI’s *An explicit power saving for the exact discrete Fourier transform* (25 September 2026). There the saving $1 - \theta$ is about 2.1×10^{-13} . Two changes give the improvement. First, a frame transition of residual dimension ρ is executed as one recursive call to $C^{\otimes \rho f}$ instead of ρ separate directional steps. Second, the fixed finite network is replaced by the complex supplier of the community repository **CrocSwap/integer-mult-bounds**. That network was built by many contributors, credited in Section 7. The rest of OpenAI’s argument is used unchanged. The same bound holds for exact complex convolution.

1 Result

We use the model of [1, §1.1] without change: exact complex arithmetic at unit cost, input-independent scalars, a specified Fourier root, and unit-cost integer operations on $O(\log(n+2))$ -bit words. Scalar preparation, schedule construction, index work and data movement are charged. Throughout,

$$C = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$$

is the matrix of [1, (2.1)].

Theorem 1.1. *There is a deterministic algorithm that, for every $n \geq 1$ and every $x \in \mathbb{C}^n$, computes $F_n x$ exactly in $O(n(\log n)^\theta (\log \log n)^{4-\theta})$ operations, where $\theta = 1 - 607/1250000$. The same bound holds for the linear convolution of two vectors in $\mathbb{C}^n$.*

Corollary 1.2. *For every fixed $\kappa < 607/1250000$, both problems take $O(n(\log n)^{1-\kappa})$ operations; for example, $\kappa = 4.85 \times 10^{-4} > 2^{-11.01}$.*

In [1] the exponent comes from one fixed network through [1, Theorem 2.6]: $C^{\otimes k}$ costs $O(2^k(k+1)^{\theta_0})$ with $\theta_0 = \log_m \lambda$ and $\lambda = m - \Delta/W_*$. In that recurrence every unit of residual dimension is a separate directional step on a fiber of f bits, so it pays one full level of recursion. Here a residual of dimension ρ is a single call on ρf bits (Theorem 3.1). The critical exponent then solves the moment equation $\sum_\rho N_\rho (\rho/m)^\theta = W$. The same batched accounting is used for integer multiplication in [2], where it was introduced by icekylinx (“projector batching” and “whole-residual complex recursion”). Section 5 explains why the remaining sections of [1] apply unchanged.

2 Batched networks

Write an array address as f columns of m bits, followed by spectator bits. A *columnwise* map applies the same map to every column.

Definition 2.1. A *batched network* of width m on W roles is a finite list of operations on W role arrays and finitely many temporary arrays. It must have the following property for every $f \geq 1$ and every spectator set: for arbitrary initial role arrays $A_1, \dots, A_W$, with temporaries starting at zero, the list leaves every role array equal to $C^{\otimes mf} A_w$, tensored with the identity on spectators. Each operation is one of the following.

1. A *gate*: a fixed linear map with coefficients in $\mathbb{Q}(i)$, applied pointwise across a fixed set of arrays (the same address in each array). Copying an array, erasing a temporary and renaming arrays are gates.
2. An *adapter* on one array: a columnwise invertible affine binary address map, followed by multiplication by $u i^{q(x)}$, where q is a sum over columns of one fixed function $\mathbb{F}_2^m \rightarrow \mathbb{Z}/4$ and $u \in \{\pm 1, \pm i\}$.
3. A *child* of width $\rho \in \{1, \dots, m\}$ on one array: $C^{\otimes \rho}$ or $C^{-\otimes \rho}$ on ρ fixed bit positions of every column, that is, $C^{\pm \otimes \rho f}$ along those ρf bits.

Let N_ρ be the number of children of width ρ .

Children on temporaries are counted in N_ρ , but temporaries are not roles. This is the usual rank deficit: $\sum_\rho \rho N_\rho < Wm$ is necessary for a saving, and copies of shared values spend part of it.

Lemma 2.2. Let $k = mf + r$ with $0 \leq r < m$. Fix an adapter as in Definition 2.1 and a set of ρ bit positions in a column. Applying the adapter to an array of length 2^k , and then moving the ρ selected bits of every column into one contiguous least-significant field of ρf bits, takes $O(2^k)$ time. This includes index preparation and data movement. The constant depends on m and the fixed maps, but not on f or k . The inverse movement has the same bound.

Proof. This is the proof of [1, Lemma 2.5] with two additions. For every column digit, the finite table also records the selected ρ bits and the value of the phase function on that column. The prefix traversal carries a third partial value, the accumulated phase in $\{\pm 1, \pm i\}$. Extending a prefix by a digit multiplies it by the tabulated phase in constant time. At a leaf the value is copied between the two computed addresses and multiplied by the accumulated phase and by u . The selected field has ρf bits, so each fiber of the following child is contiguous, and W consecutive fibers form a contiguous block of $W2^{\rho f}$ entries. $\square$

3 The batched recursion

Theorem 3.1. Let $\mathcal{N}$ be a batched network of width m on W roles whose children all have width at most $m - 1$. If $0 < \theta < 1$ and

$$\mu := \frac{1}{W} \sum_\rho N_\rho \left(\frac{\rho}{m}\right)^\theta < 1,$$

then $C^{\otimes k}$ can be applied exactly to an arbitrary complex array of length 2^k in $O(2^k(k+1)^\theta)$ time, for every $k \geq 0$. The bound includes fixed-table construction, all scalar arithmetic, index preparation and data movement.

Proof. We describe a procedure $\mathcal{A}(k)$ that applies $C^{\otimes k}$ to each of W arbitrary arrays of length 2^k , and bound its cost $T(k)$. Put $t(k) = T(k)/(W2^k)$. Let $\rho_{\max} \leq m - 1$ be the largest child width of $\mathcal{N}$.

For $k < K$, with K fixed below, apply the k factors of C one at a time along their fibers. This costs $O(kW2^k)$, so $t(k) \leq c_1 K$.

For $k \geq K$ write $k = mf + r$ with $0 \leq r < m$. Apply the r spectator factors directly, at cost $O(rW2^k)$. Then execute $\mathcal{N}$ on the W arrays with f columns. Gates cost $O(2^k)$ per array touched. Adapters cost $O(2^k)$ by Lemma 2.2. A child of width ρ on one array first moves its selected bits into the low field (Lemma 2.2), so that the array consists of $2^{k-\rho f}$ contiguous fibers of length $2^{\rho f}$. Group the fibers into $\lceil 2^{k-\rho f}/W \rceil$ batches of W fibers, padding the last batch with zero arrays, and run $\mathcal{A}(\rho f)$ on each batch. Then undo the movement. An inverse child is a forward child followed by a translation [1, Lemma 2.2]. Correctness follows by induction on k , because $\rho f < k$ and Definition 2.1 holds for arbitrary role values. Zero padding is mapped to zero.

There is a constant A_0 , depending only on $\mathcal{N}$ and m , with

$$T(k) \leq A_0 W 2^k + \sum_{\text{children } c} \left(\frac{2^{k-\rho_c f}}{W} + 1 \right) T(\rho_c f).$$

Since $k - \rho_c f \geq (m - \rho_{\max})f \geq f$, dividing by $W2^k$ gives

$$t(k) \leq A_0 + (1 + W2^{-f}) \sum_{\rho} \frac{N_{\rho}}{W} t(\rho f).$$

Suppose $t(j) \leq B(j+1)^{\theta}$ for all $j < k$. Since $\rho f + 1 \leq (\rho/m)(k+1) + 1 \leq (\rho/m)(k+1)(1 + m/(k+1))$,

$$t(k) \leq A_0 + B(k+1)^{\theta} \mu (1 + W2^{-f}) \left(1 + \frac{m}{k+1} \right).$$

Choose K so large that the product of μ and the last two factors is at most $\mu' = (1 + \mu)/2 < 1$ for all $k \geq K$, and put $B = \max\{c_1 K, A_0/(1 - \mu')\}$. Then $t(k) \leq B(k+1)^{\theta}$ for every k . Finally, place the input in one role, set the others to zero, and run $\mathcal{A}(k)$. Since W is a constant, the cost is $O(2^k(k+1)^{\theta})$. Index widths, the recursion stack and workspace are bounded exactly as in the proof of [1, Theorem 2.6]: the array lengths shrink geometrically, and addresses have $k + O(1)$ bits. $\square$

Remark 3.2. [1, Theorem 2.6] is the case where every child has width one: then $\mu = S/(Wm^{\theta})$ and the condition reads $\theta > \log_m(S/W)$. In Theorem 3.1 a child of width ρ costs $(\rho/m)^{\theta}$ of a role, not $\rho(1/m)^{\theta}$. To first order in $1 - \theta$, the critical saving is the rank deficit $Wm - \sum_{\rho} \rho N_{\rho}$ divided by $\sum_{\rho} N_{\rho} \rho \ln(m/\rho)$, rather than by $\ln m \sum_{\rho} \rho N_{\rho}$. Power-of-two padding of the role count is not needed, because partial batches are lower order. Theorem 3.1 also requires the network property for arbitrary role values, for the reason given in [1, Remark 2.7].

4 The network

Proposition 4.1. *The paired-cube complex supplier of [2] at commit `d1d6c07` is a batched network of width $m = 72$ on $W = |G| \cdot 29937$ roles, with $G = O(72, 2)$ and $N_{\rho} = |G| \cdot n_{\rho}$. Here n_{ρ} is the per-vertex child histogram in `certificates/network-children.json`. Every child has width at most 60.*

Proof (by reference). The construction and its proofs are in the notes of [2], pinned at commit `d1d6c070f5a8c684727ee7ec35d930f9ebfa9758`. We use four facts, all exact identities over $\mathbb{Q}(i)$ on arbitrary role values.

1. The completed three-stage word with shared completed cores applies the exact $C^{\otimes m}$ to every persistent role, up to fixed rank-zero adapters, signs, a fixed Pauli translation and the data-bank exchange. It restores arbitrary auxiliary values. Sources: `notes/paired-cube-construction.tex` (the identity $K + H + B = I$, the dirty-word contract, copied centers), `notes/paired-cube-sharing.tex` (completed-core sharing, the raw operator contract and final gauge child), and `notes/three-stage-cover-complex.tex` (chronology and phase endpoints).
2. Every frame transition of Fourier rank ρ is two rank-zero adapters around one coordinate child $C^{\pm\otimes\rho}$. Rank-zero adapters are binary address permutations with fourth-root quadratic phases. Sources: `notes/general-clifford-frames.tex` (exact representatives, nested children, tails and complements) and the general Gauss normal form of `notes/endpoint-gauge-complex.tex`, used there for every nondegenerate residual.
3. Copied centers are temporary streams: copy, transform, read, erase (`notes/copied-centers-lemma.tex`). Their transforms are children.
4. The complete child multiset per cover vertex is the ledger of `notes/paired-cube-sharing.tex`, §“The complete paid ledger”, instantiated in `certificates/paired-cube-complex-input.json`: $\sum_{\rho} \rho n_{\rho} = 2153528 = 72 \cdot 29937 - 1936$, with largest child 60.

All of these are identities between Clifford operators on $\mathbb{F}_2^m$ and pointwise scalar maps across roles. Scalar maps commute with tensoring along columns, so the identities hold columnwise for f columns, with frames $\Phi^{\otimes f}$ as in [1, (2.6)]. Routing between invocations renames arrays. Every multiplicity is $|G|$ times the per-vertex count, so $|G|$ cancels from μ . $\square$

Proposition 4.2. *For the network of Proposition 4.1 and $\theta = 1 - 607/1250000$,*

$$\sum_{\rho} n_{\rho} \left(\frac{\rho}{72} \right)^{\theta} < 29937 - 0.00315,$$

so $\mu < 1$.

Proof. Write $(\rho/m)^{\theta} = (\rho/m) \exp(a \ln(m/\rho))$ with $a = 607/1250000$. The script `scripts/verify_moment.py` bounds $\ln(m/\rho)$ above by the series $2 \sum y^{2j+1}/(2j+1)$, $y = (x-1)/(x+1)$, with an explicit geometric tail. It bounds $e^x \leq 1 + x + x^2/(2(1-x/3))$ for $0 \leq x < 3$. The sum is then evaluated in exact rational arithmetic. The certified margin is recorded in `certificates/moment.json`. The critical saving of this histogram is about 4.85657×10^{-4} . As a control, the same script fails at $a = 4.8566 \times 10^{-4}$. $\square$

5 Proof of Theorem 1.1

By Propositions 4.1 and 4.2, Theorem 3.1 gives $C^{\otimes k}$ in $O(2^k(k+1)^{\theta})$ with $\theta = 1 - 607/1250000$. The remaining sections of [1] use their Theorem 2.6 only through this bound. That covers the exact-width Fourier words of §3, the sector synchronization of §4 (Proposition 4.2, where each width- 2^k sector carries $C^{\otimes k}$), and the small-prime working lengths and chirp convolution of §5. Their final estimate also uses $\theta > 0$. With our θ their proof of Theorem 1.1 gives $O(n\ell^{\theta}(\log \ell)^4)$, where $\ell = \Theta(\log n / \log \log n)$ is the axis count. This is $O(n(\log n)^{\theta}(\log \log n)^{4-\theta})$. Their proof of Corollary 1.3 uses three transforms of length $2n - 1$, which gives the convolution bound.

The fixed network enters only through finite tables and constants. Its role count $W \approx 2^{2446}$ and the base range K are absolute constants, as in [1]. They are astronomically large: this is an asymptotic existence and explicitness result, with no practical crossover.

6 Scope

This note is a paper proof plus an exact finite certificate. It is not formally verified; [1] has a Lean formalization that does not cover the changes here. The network’s exact identities are proved in the cited notes of [2], which check them by scripts and maintainer review. In [2] the network is one input to a conditional integer-multiplication bound. The conditional hypotheses there concern that repository’s analytic and tape assembly; none of them is used here.

7 Credits

The finite network is community work recorded in [2]. Its NOTICE, CONTRIBUTORS file and docs/research/community-round6-review.md at commit d1d6c07 give the authoritative attribution; all of it applies here. The principal contributions to this network are:

- **icekylinx** (with OpenAI GPT-6 Astra and Codex assistance): the paired-cube identity $K + H + B = I$, signed overlap channels, the original-source involution, coordinate-star centers, the frame/carrier compiler and chronological gauges; the arbitrary-subspace Clifford frame theorem and the three-stage Cayley cover; copied retained centers; endpoint gauges; and the batched (whole-residual) recursion that Theorem 3.1 adapts.
- **an664** (with OpenAI Codex assistance): completed-core workspace sharing (PR #128).
- **eumemic** (with Claude assistance): the positive producer DAG whose restrictions supply the coarse query modules (PR #117), and earlier compressed complex networks and source frames (PRs #3, #13).
- **Aurel Prosz (Paureel)**: the copy/transform/read/discard interface and the two-stage topology. **Zhihao Chen (jacklightChen)**: translated complex endpoints and two-stage composition. **Swapnil Jain**: two-stage development. **dleen**: retained totals and bank sharing in the complex network (PR #4).
- **Douglas Colkitt**: the repository, its compact-control framework, integration and review.
- **OpenAI**: the original complex phase network of *Integer multiplication below $n \log n$* , and the exact-transform framework of [1] reused here.

Other contributors credited in [2] for parts of this network’s lineage retain that credit. This note claims only the transfer of batched accounting to the exact-DFT model (Lemma 2.2, Theorem 3.1), the certificate of Proposition 4.2, and the observation that the community network applies.

References

- [1] OpenAI, *An explicit power saving for the exact discrete Fourier transform*, 25 September 2026. Repository <https://github.com/openai/math>, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, directory `preprints/An-explicit-power-saving-for-the-exact-discrete-Fourier-transform-September-25-2026`.
- [2] D. Colkitt et al., *A sharper exponent for integer multiplication* (community repository), <https://github.com/CrocSwap/integer-mult-bounds>, commit `d1d6c070f5a8c684727ee7ec35d930f9ebfa9758`.